

BM4Midpoint: Arithmetic Diagonal Projection

GC2D numerical methods

1 Complete-cycle definition

For $\dot{z} = f(t, z)$, embed the accepted state as $Ez = (z, z)$ and define $P(u, v) = (u + v)/2$. One step is

$$z_{n+1} = P\Psi_h(Ez_n),$$

where Ψ_h is the same unprojected twelve-stage BM4 composition used by BM4Implicit. No projection is applied between stages and no multiplier is solved. “Midpoint” refers to the final arithmetic mean.

The coefficient vector is $(b_1, \dots, b_6, b_6, \dots, b_1)$, where

$$\begin{aligned} b_1 &= 0.0792036964311957, & b_2 &= 0.1303114101821663, \\ b_3 &= 0.2228614958676077, & b_4 &= -0.3667132690474257, \\ b_5 &= 0.3246481886897062, & b_6 &= 0.1096884778767498. \end{aligned}$$

The sum is one to floating-point accuracy. Starting with the adjoint, stages alternate adjoint and direct maps with signed duration $d_j = b_j h$. If $s_j = t_n + \sum_{i < j} d_i$, an adjoint stage uses s_j and a direct stage uses $s_j + d_j$. The negative coefficient also moves the stage clock backwards.

At the chosen evaluation time, let

$$A_d(u, v) = (u, v + df(t, u)), \quad B_d(u, v) = (u + df(t, v), v).$$

Let C_d be the exact GC harmonic mixing flow: it preserves the copy mean and rotates the spatial difference through angle $2\omega d$. The direct map is $C_d \circ B_d \circ A_d$; its adjoint is $B_d \circ A_d \circ C_d$. Each stage evaluates the vector field twice. Thus one complete step uses 24 evaluations and one arithmetic projection.

2 State and energy conventions

Physical mode packs p particles as $(x_1, \dots, x_p, y_1, \dots, y_p)$. Its accepted state has dimension $2p$ and its temporary copies have dimension $4p$. Optional energy tracking appends p auxiliary momenta. Each shear adds $d(-\partial_t H)$ to the summed momentum k at that shear’s evaluation state. The reported physical momentum is $\kappa = k/2$, initialized at zero. These updates are triangular and do not affect the physical trajectory.

3 Fully extended state

As in ABBA2Midpoint, fully extended mode supports one GC particle and uses

$$Z = (x, y, t, k), \quad X_K(Z) = (f(t, x, y), 1, -\partial_t H(t, x, y)).$$

Both copies contain all four coordinates. Replace f in the two shears by X_K , evaluate at each copy's intrinsic time, and apply C_d only to the spatial coordinates. The same twelve-stage traversal is followed by $Z_{n+1} = P\Psi_h(EZ_n)$. Time advances analytically by h ; the implementation checks and synchronizes it after each complete step.

Energy tracking is inherent. The reported momentum is the averaged intrinsic k , without the physical-mode factor of two. The diagnostic generalized energy is $H(t, x, y) + k$. Public solution states contain only (x, y) ; observers receive the accepted four-dimensional map, including perturbations of time and momentum.

4 Order and geometric limitations

On the invariant diagonal of the exact extended flow, the two copies follow the original dynamics and the harmonic coupling vanishes. For a smooth field, the fourth-order unprojected composition therefore satisfies

$$\Psi_h(Ez) = E\varphi_h(z) + O(h^5).$$

Since $PE = I$ and P is linear, the projected method has local error $O(h^5)$ and global order four under the usual stability assumptions. The same argument applies to the autonomous fully extended field.

Arithmetic projection discards the copy difference. Consequently, symmetry or symplecticity of a base composition alone does not establish exact reversibility or symplecticity of the accepted physical map. This method makes neither claim. The recorded infinity norm $\|u_f - v_f\|_\infty$ measures the discarded separation before projection.

5 Executable checks

The test suite compares the physical map with an independent signed-stage replay and checks fourth-order convergence against an exact rotation and a tightly integrated nonautonomous reference for both state extensions. It also checks momentum normalization against an augmented reference ODE, particle batching, persistent observer callbacks, and independence from off-grid output sampling. Nonlinear iterations are absent by construction.